

PriceWise

Material Price Intelligence & Procurement Decision Engine

FABathon 2026 Solution

Tata Electronics / TSMPL Semiconductor Sourcing Challenge

All data is synthetic — demo purposes only



1. The Problem: Semiconductor Sourcing Complexity

Semiconductor manufacturing relies on highly specialized consumables and raw materials (CMP Slurries, Photoresists, Sputtering Targets, High-Purity Gases).

Key Sourcing Challenges:

- **Limited Historical Data:** Sparse price histories for custom formulations.
- **Fragmented Market Intelligence:** Unstructured news, geopolitical events, and tariff changes.
- **Expertise Bottleneck:** Sourcing decisions rely on tacit buyer knowledge rather than systemized intelligence.
- **Black-Box ML Rejection:** Procurement teams distrust purely statistical "black-box" predictions.

2. Core Product Philosophy: 4 Key Sourcing Questions

PriceWise is an **explainable decision-support engine**, not a black-box forecaster.

1. WHAT is likely to happen?

Directional point forecasts with confidence bounds and time horizons.

2. WHY is it likely to happen?

Decomposition into real economic cost drivers (Rare Earths, Energy, Freight, FX).

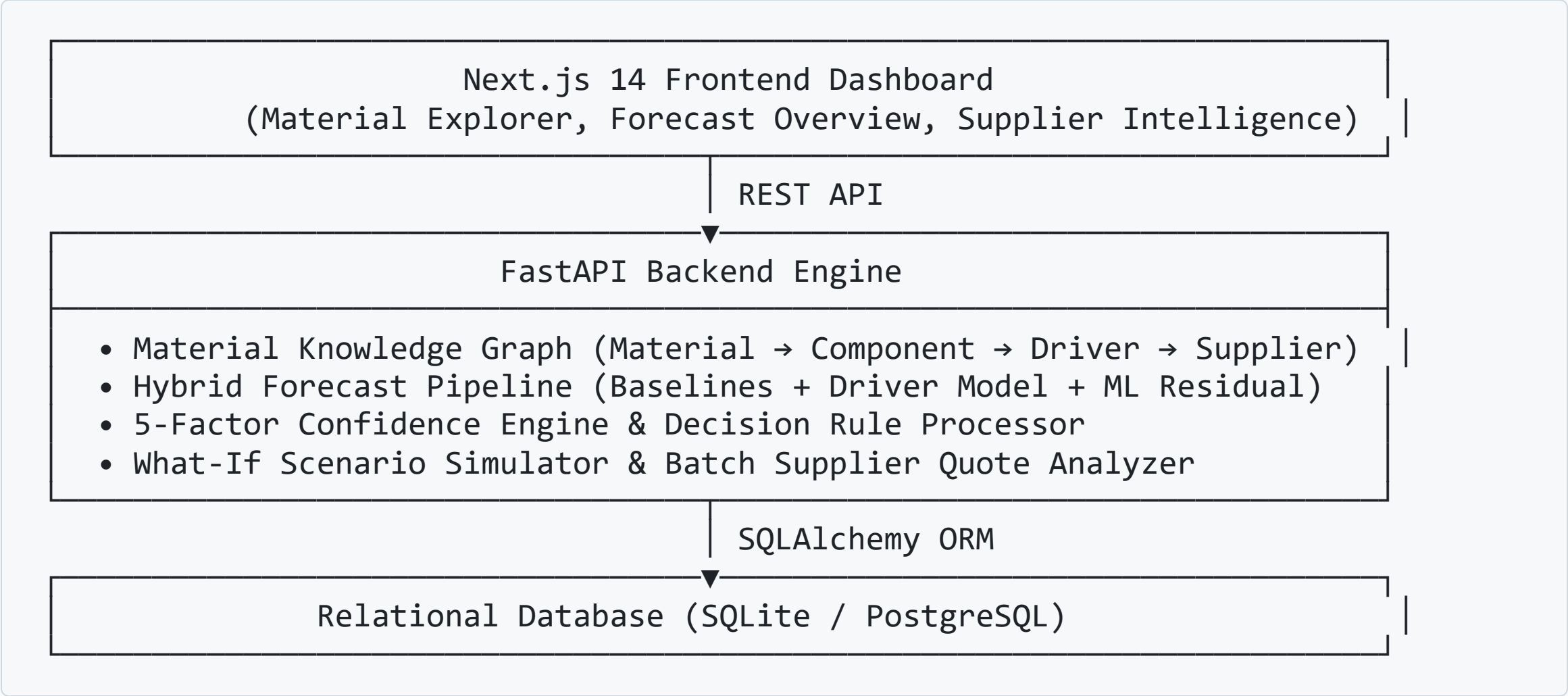
3. HOW CONFIDENT are we?

Transparent 5-component confidence scoring heuristic (0–100 score).

4. WHAT SHOULD PROCUREMENT DO?

Actionable strategies (`SHORT\_LOCK`, `NEGOTIATE`, `DUAL\_SOURCE`, `WAIT`).

3. End-to-End System Architecture



4. The Material Knowledge Graph

PriceWise structures materials as a **hierarchical graph/tree** rather than flat text.

```
CMP Slurry (Material)
├── Ceria Abrasive (Component - 42% cost)
│   ├── Rare Earth Index (Driver) — [Strength: 0.87]
│   └── FX Index (Driver) — [Strength: 0.30]
├── Oxidizer / H2O2 (Component - 18% cost)
│   └── Energy / Industrial Gas — [Strength: 0.65]
└── Logistics & Dispersant (Component - 40% cost)
    └── Freight Index — [Strength: 0.50]
```

- Enables economics-grounded price modeling.
- Driver relationships are weighted by **component cost share × elasticity**.

5. Hybrid Forecasting Engine

To handle limited semiconductor time-series data without overfitting, PriceWise uses a 4-tier hybrid pipeline:

1. **Statistical Baselines:** Moving Average, Exponential Smoothing, Seasonal Naive.
2. **Economically Interpretable Driver Model:** Ridge regression mapped to knowledge graph weighted drivers:
$$\text{Price Change} = \text{Intercept} + \sum(\text{Weight}_i * \text{Driver}_i \text{ Change})$$
3. **ML Residual Model:** LightGBM trained on residual errors ($\text{Actual Price Change} - \text{Driver Forecast}$). Skipped automatically when history < 18 months.
4. **Walk-Forward Ensemble:** Time-series validation weights models based on historical accuracy without data leakage.

6. Adaptive Data Modes & Explainability

Adaptive Low-Data Handling

- **Fewer than 12 observations (`LOW_DATA`):** Suppresses ML model; relies on baseline + driver model; reduces confidence score automatically.
- **More than 48 observations (`STRONG_DATA`):** Full ensemble enabled with ML residual modeling.

Driver Contribution Waterfall

- Allocates price changes across real economic drivers (e.g., Ceria +4.1%, Energy +1.2%, Freight +0.8%, FX +0.7%).
- Generates plain-English narrative explanations without relying on external LLM calls.

7. 5-Factor Confidence Scoring Framework

PriceWise computes a transparent, explainable confidence score (0 to 100):

$$\text{Overall Score} = 0.20 \times \text{Data} + 0.25 \times \text{Driver} + 0.25 \times \text{Model} + 0.15 \times \text{Market} + 0.15 \times \text{Stability}$$

Data Quality (20%)

Evaluates history length, sample frequency, and data mode tier.

Driver Strength (25%)

Aggregates knowledge graph edge relationship strengths.

Model Performance (25%)

Measures walk-forward backtest directional accuracy and MAPE.

Stability & Signals (30%)

Penalizes model disagreement and sudden volatility regime shifts.

8. Procurement Decision Engine

Translates quantitative forecast outputs and supply risk into actionable sourcing guidance.

Decision Rule Logic

- Rising Forecast + High Confidence + High Supply Risk: → **SHORT_LOCK** (3 Months)
- Rising Forecast + High Confidence + Manageable Risk: → **NEGOTIATE** (1–3 Months)
- Falling Forecast + High Confidence + Manageable Risk: → **WAIT** (1 Month)
- High Supplier Concentration / Single Source: → **DUAL_SOURCE**

Includes an **Evidence & Audit Trail** linking every recommendation back to forecast bounds, market events, and supplier risk factors.

9. Integrated Supplier Intelligence & Quote Analysis

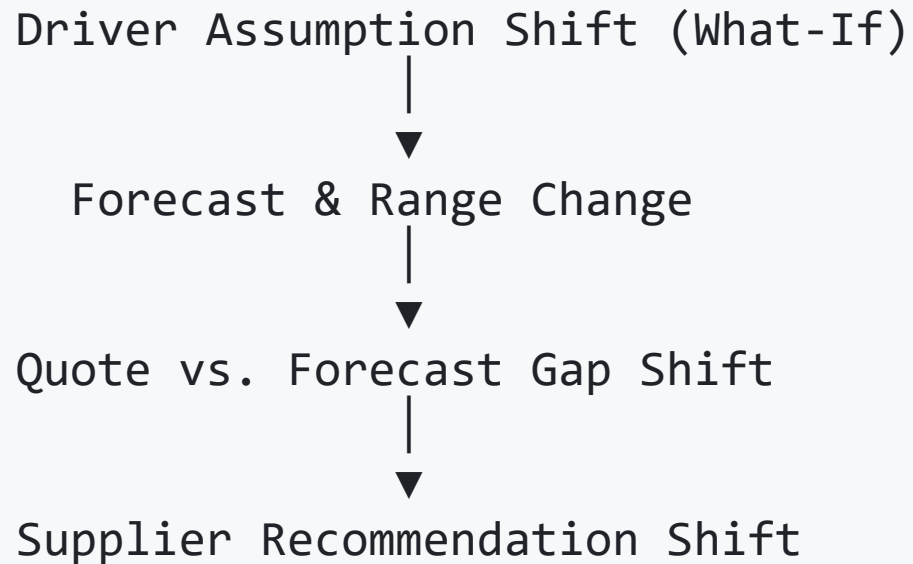
Converts static supplier lists into an active decision-support system.

Core Functionality

- **Supplier Catalog & Detail View:** Qualification status, lead time, share of supply, risk scores, and quote history.
- **Dynamic Quote Batch Analysis:** Compares supplier quotes against active market forecast.
- **Claim Support & Negotiation Anchor:**
 - Identifies **Market-Supported Change** vs. **Unexplained Premium**.
 - Classifies quotes as **SUPPORTED**, **PARTIALLY_SUPPORTED**, or **UNSUPPORTED**.
 - Recommends specific negotiation target prices.

10. What-If Scenario Simulator & Causal Chain

Allows procurement managers to simulate driver shocks (e.g., Rare Earth +15%, Freight +10%) and observe real-time causal propagation:



Example: A supplier quote deemed **UNSUPPORTED** under base market conditions may become **SUPPORTED** under an elevated energy/driver scenario.

11. Demonstration Capabilities (Seeded Hero Story)

PriceWise includes a complete seed pipeline with synthetic semiconductor materials:

- **MAT-001 (Ceria CMP Slurry):** 48-month history, single-source, hero narrative +9% supplier quote claim vs +6.2% market forecast.
- **MAT-002 (Hydrogen Peroxide):** Commodity chemical driven by energy indices.
- **MAT-003 (Specialty Photoresist Polymer):** 8-month sparse history showcasing automatic `LOW_DATA` mode and reduced confidence.
- **MAT-004 (Copper Sputtering Target):** Metal-index driven material.

12. Future Roadmap & Scaling Vision

Phase	Horizon	Focus Areas
Phase 4	Market Intelligence	LLM-driven unstructured news/event extraction, impact scoring, & source credibility rating.
Phase 5	Live Connectors	Integration with live commodity feeds (LME, Platts, ICIS) & ERP systems (SAP/Oracle).
Phase 6	Advanced Sourcing	Automated supplier negotiation playbook generator & multi-tier BOM risk modeling.

Thank You!

PriceWise

*Material Price Intelligence & Procurement Decision
Engine*

Empowering semiconductor supply chains with explainable, data-driven sourcing.

